

LAGOS IKEJA, Nigeria

WMO: 652010

Lat: 6.577N

Lon: 3.321E

Elev: 135

StdP: 14.62

Time Zone: 1.00 (E01)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.9	71.5	51.4	56.6	85.5	57.4	70.7	83.1	22.3	81.1	20.7	81.5	3.1	0	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	14.2	94.7	78.6	93.4	79.1	92.7	79.1	83.0	89.6	82.2	89.1	81.7	88.3	7.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.8	86.2	80.6	159.9	85.8	80.0	156.6	85.3	47.1	90.0	46.1	89.6	45.5	89.0	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.3	17.4	14.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	83.0	84.5	85.2	84.2	82.7	80.3	79.2	78.8	79.4	80.6	82.8	83.3
	DBStd	2.92	2.47	2.31	1.87	2.49	2.27	2.10	1.62	1.40	1.50	1.96	1.69	1.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11675	1023	967	1091	1027	1013	910	905	893	883	947	983	1033
	CDD65	6200	558	547	626	577	548	460	440	428	433	482	533	568
	CDH74	61500	5824	6238	7504	6814	5934	4105	3324	2904	3262	4213	5459	5919
	CDH80	22011	2459	2663	3355	2985	2224	1059	576	446	608	1185	2023	2428
Wind	WSAvg	5.3	4.3	5.4	5.4	6.2	5.0	5.1	6.2	7.0	5.7	4.7	4.0	4.1
Precipitation	PrecAvg	61.8	0.6	1.5	3.1	6.3	8.1	12.5	9.5	4.1	7.0	5.9	2.2	0.7
	PrecMax	103.0	2.4	7.4	9.8	15.3	17.7	28.8	23.0	18.4	16.5	12.8	5.4	3.9
	PrecMin	34.3	0.0	0.0	0.2	0.4	1.4	4.6	0.9	0.1	1.1	1.5	0.0	0.0
	PrecStd	13.9	0.6	1.6	2.0	3.1	3.4	5.6	6.5	4.3	3.9	2.5	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.4	96.2	95.1	95.0	93.2	90.5	87.6	86.6	88.1	90.5	93.5	95.3
		MCWB	75.9	77.2	80.1	81.3	81.2	80.1	78.5	76.9	78.9	80.7	78.5	76.7
	2%	DB	93.5	93.6	93.5	93.3	91.5	88.2	86.0	85.4	86.3	89.2	91.7	93.5
		MCWB	76.4	79.6	80.8	81.2	80.9	79.7	78.1	76.7	78.1	80.0	79.4	76.6
	5%	DB	92.3	92.8	92.7	91.8	90.0	86.8	84.5	84.1	84.8	87.6	90.1	92.0
		MCWB	76.2	79.7	80.7	80.7	80.3	79.2	77.4	76.5	77.6	79.6	79.7	77.1
	10%	DB	90.7	91.0	91.3	91.0	89.2	85.7	83.0	82.5	83.9	86.1	89.1	90.5
		MCWB	75.8	79.6	80.3	80.4	80.1	78.8	76.8	75.8	77.3	79.0	79.8	77.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.2	83.7	84.2	83.6	82.8	81.4	79.7	78.6	80.0	81.9	82.8	83.5
		MCDB	89.0	89.5	90.7	90.9	89.3	86.7	84.6	83.3	85.2	87.6	89.8	89.3
	2%	WB	81.4	82.1	82.6	82.4	81.7	80.4	78.8	77.6	79.1	80.7	81.8	81.9
		MCDB	88.2	89.0	89.7	89.5	88.4	86.0	83.6	82.5	84.1	86.9	88.5	88.5
	5%	WB	80.3	81.3	81.8	81.7	81.0	79.7	78.2	77.1	78.3	79.9	81.2	80.9
		MCDB	86.7	87.7	88.5	88.3	87.4	84.9	82.6	82.0	83.0	85.6	87.7	87.6
	10%	WB	79.6	80.7	81.0	81.1	80.4	79.1	77.6	76.5	77.7	79.1	80.4	80.1
		MCDB	85.4	86.7	87.3	87.3	86.5	83.8	81.7	80.9	82.1	84.3	86.5	86.1
Mean Daily Temperature Range	5% DB	MDBR	18.1	15.0	14.2	14.1	13.3	10.9	9.2	9.1	10.0	12.5	14.8	17.1
		MCDBR	19.1	15.8	14.9	14.9	14.4	12.9	11.0	10.8	11.7	14.1	15.9	18.6
		MCWBR	7.5	5.2	5.6	5.6	6.0	6.1	4.6	4.5	5.3	6.7	6.9	6.9
	5% WB	MCDBR	16.2	14.5	14.0	14.1	13.7	11.9	9.9	9.5	10.7	13.2	14.7	15.6
		MCWBR	7.0	6.2	6.2	5.9	6.5	6.2	4.7	4.4	5.3	6.9	7.1	6.8
Clear-Sky Solar Irradiance	taub	0.844	0.852	0.833	0.683	0.572	0.535	0.531	0.538	0.493	0.524	0.622	0.728	
	taud	1.481	1.481	1.499	1.762	1.986	2.101	2.114	2.084	2.237	2.134	1.844	1.654	
	Ebn at Noon	179	183	189	217	236	242	244	247	262	254	226	199	
	Edh at Noon	96	98	97	74	57	50	50	53	46	50	66	80	
All-Sky Solar Radiation	RadAvg	1559	1537	1602	1619	1485	1238	1214	1265	1315	1470	1590	1561	
	RadStd	63	94	81	68	58	97	115	82	81	56	54	55	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.50	+0.56	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page